Pipeline: Original Graph G → Generate Node Features X → Construct Graph Pair (Gs, Gt) + Ground Truth → Add Noise (Structural + Attributive) → Convert to Matrix Form (A, X) → Embedding / Representation Learning → Compute Similarity Matrix → Matching (Alignment) → Evaluation (Accuracy / Hit@k)

1. Environment Setup

```
!pip  install  "numpy<2"  node2vec
!pip  install  scikit-learn  scipy  pandas  matplotlib  torch  networkx
```

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Successfully installed contourpy-1.3.2 cycler-0.12.1 filelock-3.28.0 fonttools-4.62.1 fsspec-2026.3.0 kiwisolver-1.5.0 matplotlib-3.10.8 mpmath-1.3.0 pillow-12.2.0 pyparsing-3.3.2 scikit-learn-1.7.2 sympy-1.14.0 threadpoolctl-3.6.0 torch-2.11.0
```

2. Path Configuration and Import

```
import  os
import  sys
sys.path.append(os.path.abspath("src"))

from  data_utils  import  *
from  noise_utils  import  *
from  matching  import  *
from  metrics  import  *
from  baselines  import  *
from  models  import  *
from  experiment  import  *
```

```
C:\Users\23293\anaconda3\envs\netalign\lib\site-packages\tqdm\auto.py:21: TqdmWarning: IProgress not found. Please update jupyter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
```

3. Parameter Configuration

```
SEED  =  42
DEBUG  =  True

EDGE_NOISE_LEVELS  =  [0.0,  0.1,  0.2,  0.3,  0.5]
ATTR_NOISE_LEVELS  =  [0.0,  0.1,  0.2,  0.3,  0.5]

EMBED_DIM  =  64
NUM_RUNS  =  3
TOPK  =  [1,  5]
USE_HUNGARIAN  =  True
DEVICE  =  "cuda"   #  or  "cpu"

DATASET_NAME  =  "karate"
```

4. Debugging a Single Run

```
set_seed(SEED)

G,  X  =  prepare_dataset(DATASET_NAME,  feat_dim=16)
Gs,  Gt,  Xs,  Xt,  gt_map  =  generate_permuted_graph_pair(G,  X)
Gs_n,  Gt_n,  Xs_n,  Xt_n  =  build_noisy_pair(Gs,  Gt,  Xs,  Xt,  edge_noise=0.1,  attr_noise=0.2)

As,  src_nodes  =  graph_to_adjacency(Gs_n)
At,  tgt_nodes  =  graph_to_adjacency(Gt_n)
gt_idx  =  build_gt_index_map(src_nodes,  tgt_nodes,  gt_map)
```

5. Run the baseline

```
Zs,  _  =  node2vec_embed(Gs_n,  dim=EMBED_DIM)
Zt,  _  =  node2vec_embed(Gt_n,  dim=EMBED_DIM)
sim  =  similarity_matrix(Zs,  Zt)
pred  =  match_nearest_neighbor(sim)

print("Node2Vec  Accuracy:",  accuracy_at_1(pred,  gt_idx))
print("Node2Vec  Hit@1:",  hit_at_k(sim,  gt_idx,  1))
print("Node2Vec  Hit@5:",  hit_at_k(sim,  gt_idx,  5))
```

```
Computing transition probabilities: 100%|██████████████████████████████████████████████████| 34/34 [00:00<00:00, 2954.29it/s]
Generating walks (CPU: 1): 100%|████████████████████████████████████████████████████████████| 50/50 [00:00<00:00, 557.62it/s]
```

```
Computing transition probabilities: 100%|██████████████████████████████████████████████████| 34/34 [00:00<00:00, 5307.07it/s]
Generating walks (CPU: 1): 100%|████████████████████████████████████████████████████████████| 50/50 [00:00<00:00, 613.82it/s]Node2Vec Accuracy: 0.23529411764705882
Node2Vec Hit@1: 0.23529411764705882
Node2Vec Hit@5: 0.5294117647058824
```

Final Run:

```
final = FINALAligner(alpha=0.8, max_iter=30).fit(As, At, Xs_n, Xt_n)
S = final.predict_score_matrix()
pred_final = final.predict_top1(use_hungarian=False)

print("FINAL Accuracy:", accuracy_at_1(pred_final, gt_idx))
print("FINAL Hit@1:", hit_at_k(S, gt_idx, 1))
print("FINAL Hit@5:", hit_at_k(S, gt_idx, 5))
```

```
FINAL Accuracy: 0.058823529411764705
FINAL Hit@1: 0.058823529411764705
FINAL Hit@5: 0.17647058823529413
```

6. Running the Overall Experiment

```
df = run_experiments(
        G, X,
        edge_noise_levels=EDGE_NOISE_LEVELS,
        attr_noise_levels=ATTR_NOISE_LEVELS,
        runs=NUM_RUNS,
        embed_dim=EMBED_DIM
)

df.head()
```

```
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 5226.93it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 457.38it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 3242.97it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 60.37it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 11317.06it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 549.92it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 5487.81it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 589.84it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 4776.95it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 618.37it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 5226.55it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 631.16it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 5671.36it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 616.70it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 6169.96it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 567.49it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 6171.56it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 613.30it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 5221.76it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 628.89it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 6163.56it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 461.67it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 5224.82it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 541.81it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 6163.83it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 619.90it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 5661.45it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 596.22it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 5473.91it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 605.27it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 5667.75it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 619.38it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 6169.43it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 780.58it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 4857.16it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 520.99it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 4521.30it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 609.80it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 4232.65it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 621.46it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 4851.88it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 569.52it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 5221.38it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 538.90it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 5224.82it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 547.21it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 5374.27it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 605.70it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 6158.24it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 533.92it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 6012.83it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 531.08it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 5666.63it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 611.54it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 6227.62it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 605.17it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 3990.55it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 574.35it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 3989.77it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 583.91it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 6799.84it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 485.80it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 6806.66it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 695.52it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 4521.30it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 580.66it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 5224.25it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 660.06it/s]
```

7. Save Results

```
df.to_csv("results/experiment_results.csv", index=False)
```

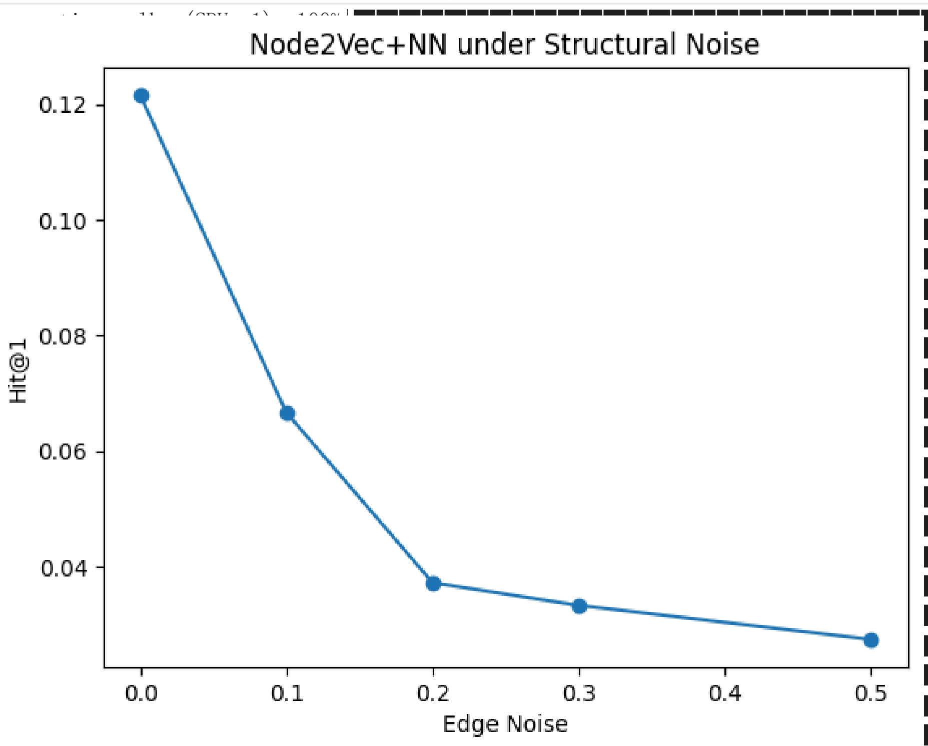
```
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 4857.16it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 558.29it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 6169.96it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 625.05it/s]
```

8. Drawing

```
import matplotlib.pyplot as plt
import pandas as pd

subset = df[df["method"] == "Node2Vec+NN"]
mean_df = subset.groupby("edge_noise")["hit1"].mean().reset_index()

plt.plot(mean_df["edge_noise"], mean_df["hit1"], marker="o")
plt.xlabel("Edge Noise")
plt.ylabel("Hit@1")
plt.title("Node2Vec+NN under Structural Noise")
plt.show()
```




```
████████████████████| 50/50 [00:00<00:00, 557.75it/s]
████████████████████| 34/34 [00:00<00:00, 6173.97it/s]
████████████████████| 50/50 [00:00<00:00, 640.41it/s]
████████████████████| 34/34 [00:00<00:00, 6609.18it/s]
████████████████████| 50/50 [00:00<00:00, 599.95it/s]
████████████████████| 34/34 [00:00<00:00, 5417.97it/s]
████████████████████| 50/50 [00:00<00:00, 616.20it/s]
████████████████████| 34/34 [00:00<00:00, 5361.34it/s]
████████████████████| 50/50 [00:00<00:00, 593.05it/s]
████████████████████| 34/34 [00:00<00:00, 6109.43it/s]
████████████████████| 50/50 [00:00<00:00, 604.97it/s]
████████████████████| 34/34 [00:00<00:00, 6165.96it/s]
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████████████████████| 34/34 [00:00<00:00, 6805.04it/s]
████████████████████| 50/50 [00:00<00:00, 637.38it/s]
████████████████████| 34/34 [00:00<00:00, 6799.84it/s]
████████████████████| 50/50 [00:00<00:00, 594.77it/s]
████████████████████| 34/34 [00:00<00:00, 5219.66it/s]
████████████████████| 50/50 [00:00<00:00, 630.15it/s]
████████████████████| 34/34 [00:00<00:00, 7525.00it/s]
████████████████████| 50/50 [00:00<00:00, 627.27it/s]
████████████████████| 34/34 [00:00<00:00, 6799.84it/s]
████████████████████| 50/50 [00:00<00:00, 633.36it/s]
████████████████████| 34/34 [00:00<00:00, 6785.61it/s]
████████████████████| 50/50 [00:00<00:00, 565.03it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 6791.75it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 627.58it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 5660.78it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 596.22it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 6164.63it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 598.71it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 6158.24it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 634.03it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 8500.11it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 687.16it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 7514.30it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 592.07it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 9338.38it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 623.10it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 8498.59it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 634.83it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 6796.60it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 588.93it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 7538.93it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 643.39it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 13588.03it/s]
Generating walks (CPU: 1): 100%|████████████████████| 50/50 [00:00<00:00, 696.91it/s]
Computing transition probabilities: 100%|████████████████████| 34/34 [00:00<00:00, 8500.11it/s]
```